

Rice: Comprehensive Overview

Executive Summary. Rice (*Oryza* spp.) is the world's most important cereal staple, supplying roughly 30% of dietary energy and 20% of protein in developing countries. Two subspecies of Asian rice (*O. sativa*) dominate cultivation: **Indica** (long-grain, tropical) and **Japonica** (short-grain, temperate, including sticky "glutinous" types). Specialty groups such as aromatic rices (e.g. basmati, jasmine) are valued for fragrance and often lie genetically closer to Japonica. Hybrid rice (inter-subspecific crosses) now accounts for 5–15% of some countries' output, exploiting heterosis for higher yield. Globally, rice production reached ~520 million tonnes (milled basis) in 2023/24 and is projected to rise to 527.6 million in 2024/25 and 541.5 million in 2025/26 (record highs). Asia produces ~90% of rice, with China and India each at ~25% of global output. Yields have been rising: between 2000 and 2020 global rice yields grew ~19%. Asia's yield was 4.8 t/ha in 2020 (up 22% since 2000), the Americas 6.5 t/ha (+56%) and Africa only 2.2 t/ha (–5%).

Rice requires warm climates and ample water. It is typically grown in **paddy** fields (flooded, puddled soil) or as rainfed upland, with cropping calendars tied to monsoons or irrigation schedules. Management practices include seedbed nursery and transplanting (common in Asia) or direct seeding. Soils are often heavy-textured and mildly acidic (optimum pH ~5.5–6.5). Nitrogen fertilizer is critical (roughly 16–18 kg N needed per tonne of grain), along with phosphorus and potassium. Growing mechanization varies: much of Asia uses power tillers, transplanting machines and combines, whereas in Africa and parts of South Asia labor remains intensive. Water management ranges from continuously flooded to **alternate wetting and drying (AWD)** to save water and reduce methane emissions.

Breeding and genetics have been key in rice's Green Revolution: semi-dwarf high-yielding varieties and now biotech/improved lines with stress-tolerance (e.g. **Sub1** flood-tolerant rice). Marker-assisted selection (MAS) is used to stack traits like disease resistance. No GM rice is commercially grown yet, though traits like Golden Rice (vitamin A) and Bt (insect-resistant) have been developed but face regulatory hurdles (e.g. the Philippines temporarily blocked Golden Rice). Major breeding targets include yield, quality, resistance to blast, blight, pests, and tolerance to submergence, drought, salinity or heat.

Harvested rice is **threshed** and milled. Paddy processing yields brown rice (husk removed) or polished white rice (bran and germ removed). White rice has a long shelf life but loses fiber, B-vitamins and minerals in polishing. **Parboiling** (soaking, steaming, drying paddy before milling) drives nutrients from bran into the endosperm and improves hardness and shelf life. Post-harvest losses (shelling, milling inefficiencies) can reach 10–30%. Proper storage (moisture ~12–13%) is essential to prevent mold and grain pests (e.g. rice weevil, flour moth); hermetic storage and periodic fumigation (phosphine) are common controls.

Nutritionally, rice is ~80% carbohydrate, ~7% protein and <1% fat. It is low in fiber when milled white, but brown (whole-grain) rice contains significant B-vitamins (thiamin, niacin, riboflavin), minerals

(Mg, P, Mn) and fiber. White rice is often enriched with thiamin, niacin and iron to offset milling losses. Rice has a relatively high glycemic index (GI), especially white and sticky rices, contributing to post-meal glucose spikes, whereas whole-grain rice is lower GI. Anti-nutrients include phytic acid (in bran) which binds Fe and Zn. Rice is naturally gluten-free but can accumulate heavy metals like arsenic, especially in flooded soils and in bran.

Culinarily, rice is extremely versatile. It is boiled or steamed as a staple food, ground into flours (for noodles, breads like rice cakes), fermented (e.g. sake, rice wine, beer), or used in snacks and desserts (mochi, puddings). Glutinous rice is used for sticky dishes and sweets. Different varieties suit different cuisines: long-grain for pilafs and biryanis, medium for paella, short-grain for sushi or risotto. Rice bran oil (from milling) is also a popular cooking oil in Asia.

****Global Production and Trade.**** Major producers in 2022/23 were India (~136 Mt) and China (~146 Mt), followed by Indonesia, Bangladesh and Vietnam (tens of millions each). In milled terms, the United States Department of Agriculture (USDA) reported ~520 Mt global production in 2023/24. Over the last two decades output expanded from ~400 Mt (2000) to over 540 Mt (projected 2026). Figure 1 illustrates production trends.

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timeline

title Global Rice Production (milled basis)

2000 : 400

2005 : 430

2010 : 490

2015 : 520

2020 : 550

2025 : 560

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Regionally, Asia dominates (e.g. India/China ~50% of world's milled output), followed by the Americas (chiefly USA/Brazil) and smaller shares in Africa, Europe and Oceania. Average yields are highest in the Americas (~6.5 t/ha) and lowest in Africa (~2.2 t/ha). Rice yields have risen faster where more inputs and irrigation exist: for example, in Asia +22% (to 4.8 t/ha) and Americas +56% (to 6.5 t/ha) over 2000–2020, whereas Africa's yield stagnated.

Major exporters (90% of global trade) are India (world's largest), Thailand, Vietnam, Pakistan, the USA, China, Myanmar and Cambodia. In 2023 India briefly banned non-basmati rice exports (then

lifted it) to calm domestic prices. The Philippines, Nigeria and the EU (with only partial self-sufficiency) are top importers. For example, **the Philippines is the world’s largest rice importer**. Rice prices (FAO All-Rice Price Index) have been volatile: after historic peaks in 2008 and 2011, prices remained relatively subdued until a modest rise in 2025. FAO reported a 2.7% rise in the rice price index in May 2026.

Agronomy. Rice grows best in warm (20–35°C), humid climates. Key agronomic factors:

- **Soil:** Deep, fertile, clayey loams are ideal for paddy culture (to retain water); pH optimum ≈5.5–6.5. Organic matter from straw/green manure is often added.
- **Water:** Rice is water-intensive. Lowland paddy typically requires continuous flooding (~1200–2000 mm/season). Alternate Wetting and Drying (AWD) is a water-saving irrigation practice that also cuts methane emissions. Upland rice (rainfed) gets 500–900 mm seasonal rain but has lower yields.
- **Planting:** Common methods are *transplanting* (seedlings raised in nurseries then manually/mechanically transplanted) or *direct-seeding* (broadcast or drill sowing seed into puddled fields). Transplanting gives higher yields and weed control, but higher labor/cost.
- **Crop Calendar:** In tropical Asia, two or three crops a year can be grown (e.g. monsoon and dry-season rice). Seedling age at transplant is often 25–35 days. Growing duration depends on variety (90–160 days).
- **Fertilizers/Inputs:** Typical nutrient uptake is ~16–18 kg N, 4–5 kg P₂O₅ and 9–10 kg K₂O per tonne of grain. Organic amendments (manure, straw incorporation) improve soil. Use of herbicides/insecticides varies; integrated pest management is encouraged.
- **Mechanization:** In industrialized production (e.g. USA, some Chinese areas), tractors, mechanical transplanters and combine harvesters are routine. In much of Asia, transplanting and harvesting remain manual or hand-held machine operations.

Genetics and Breeding. Rice breeding focuses on yield, quality and stress resistance. Notable advances include the semi-dwarf high-yield IR8 (1960s) and more recently hybrid varieties (first commercial hybrids in China 1970s). Marker-Assisted Selection (MAS) is used for traits like blast- and blight-resistance and flood-tolerance (e.g. introgressing the SUB1A gene yields “scuba rice” that survives 10–14 days underwater). No genetically engineered rice has been commercialized globally, but Golden Rice (β-carotene-producing) is near approvals in Asia, and Bt (insect-resistance) lines exist in R&D. Modern genomics and CRISPR gene-editing are enabling “climate-smart” traits (drought, heat, salinity tolerance) and nutrient fortification, though field use is limited.

Variety Group	Origin/Subspecies	Grain	Amylose	Regions/Examples
Indica	O. sativa subspecies indica	Long, slender, non-sticky	High (20–30%)	South/Southeast Asia (IR64), US long-grain

| ****Japonica**** | *O. sativa* subspecies *japonica* (temperate/tropical) | Short, round, sticky | Low/medium (0–20%) | East Asia (Koshihikari), Italy (Arborio) |

| ****Aromatic**** | Mostly Indica or Japonica | Long, slender, aromatic | Medium-low | Basmati (India/Pak), Jasmine (Thailand) |

| ****Glutinous (Sticky)**** | Primarily Japonica | Short, opaque, very sticky | 0–1% | Southeast Asia (mochi rice, Laos sticky) |

| ****Hybrid**** | Indica×Japonica crosses | Long grains, vigorous | Varies | China hybrids (Fengliangyou 4hao) |

****Post-Harvest Processing.**** Harvested rice (paddy) is first ****threshed**** then dried (to ~14% moisture). Processing steps include:

- ****Hulling/Milling:**** Removal of hull yields brown rice (bran and germ intact). Further milling/polishing removes bran/germ to produce white rice. Polishing enhances appearance and shelf stability but strips most fiber, vitamins (B₁, B₃), minerals (Fe, Zn).

- ****Parboiling:**** Fresh paddy is soaked, steamed/boiled, and dried before milling. This gelatinizes starch and drives water-soluble nutrients from bran into the endosperm. Parboiled rice (common in South Asia) has firmer texture, fewer broken grains and better retention of vitamins/minerals after milling.

- ****Dehusking and Sorting:**** Modern mills use rubber-roll hullers, aspirators and graders to remove off-types and broken grains. Bran from milling can be used as livestock feed or processed into rice bran oil.

****Storage and Shelf-Life.**** Proper storage is crucial to preserve quality. Recommended moisture is 12–14% to prevent mold or sprouting. Whole (unmilled) paddy can be stored longer; once milled, rice begins to lose nutrients over months. Major storage pests include the rice weevil (*Sitophilus oryzae*) and *Tribolium* beetles. ****Fumigation**** (e.g. phosphine gas) or hermetic (airtight) storage bags are used to control insects without chemical residues. Temperature above 25°C accelerates spoilage. Rice can last 1–2 years if dry and cool, but insect eggs and aflatoxins from molds are concerns in warm, humid conditions.

****Nutrition and Health.**** Rice is energy-dense but low-fat. ****Macronutrients:**** Cooked white rice (~150 g) supplies ~200–220 kcal, ~4–5 g protein, <0.5 g fat; the rest (~45–50 g) is starch. Brown rice provides additional fiber (~3.5 g/150 g) and nutrients. ****Micronutrients:**** Whole rice (especially red or black varieties) contains B-vitamins (thiamin, niacin, riboflavin, B₆), vitamin E and minerals (magnesium, phosphorus, manganese, zinc). White rice is often fortified with iron, folate and B-vitamins (thiamin/niacin) to prevent deficiencies in populations relying on rice. Rice is also a source of beneficial phytochemicals like antioxidants in pigmented rices. ****Glycemic Index:**** White rice has a high GI (~70–80), so large intakes can raise blood sugar levels, whereas brown rice (fiber-rich) has a moderate GI. Diets with mostly white rice are associated with higher diabetes risk than those with whole grains. ****Anti-nutrients/Contaminants:**** Brown rice contains phytic acid which can inhibit iron and zinc absorption (reduced by cooking). Rice readily accumulates inorganic arsenic (a

carcinogen) from soil/water; bran has the highest concentration, so brown rice can contain more arsenic than polished rice. Overall, rice is gluten-free and hypoallergenic, but as a near-exclusive staple it can lead to micronutrient deficiencies unless diets are diverse or the rice is fortified.

****Food Uses and Culinary Types.**** Beyond boiled rice, rice is made into flour (for noodles, flatbreads, thickeners), puffed snacks, and fermented products (rice vinegar, wines like sake or arrack, beers like chhaang). Rice bran oil is a popular cooking oil with a high smoke point and beneficial unsaturated fat profile. Rice byproducts (straw, husk) have uses in fodder, thatch, mushroom substrate or bioenergy.

****Value Chain and Economics.**** Rice is typically sold by weight (often paddy to millers, milled to traders). Farmgate prices vary widely by variety and country. Major costs at farm level include seed, fertilizer, water and labor/energy (tilling, transplanting, harvesting). In Asia smallholders often see slim margins, relying on high yields. Wholesale rice prices are much lower in origin countries (e.g. India, Thailand) than retail prices in importing markets (especially Africa) due to tariffs, transport and packaging costs. Top exporters (India, Thailand, Vietnam) compete largely on price; quality premium rices (Aromatic basmati, Thai jasmine) fetch higher prices. Global rice price trends have been shaped by policy shocks: for example, India (~40% of world exports) imposed export curbs in 2023 to protect domestic supply, pushing world prices up. FAO's All Rice Price Index, which was low through 2020–2022, rose in 2023–24 (e.g. +2.7% in May 2026) as fuel costs and weather concerns tightened supplies.

****Sustainability & Environment.**** Rice has a large environmental footprint. Flooded paddy fields generate ****methane****, a potent greenhouse gas: globally about 8% of anthropogenic methane comes from rice cultivation. Methane arises from anaerobic decomposition of organic matter in submerged soils. Continuous flooding also uses vast water: typical water footprints are thousands of liters per kilogram of grain (making rice one of the thirstiest cereals). Alternate Wetting and Drying (AWD) irrigation can cut both water use and methane emissions by 30–50% without yield loss. Nutrient runoff (especially N and P) can pollute waterways, and straw burning in the field (common in South Asia) causes air pollution. Research is focusing on low-methane rice strains, improved irrigation, and alternative practices (e.g. direct seeding) for sustainability. Rice cultivation is also carbon-intensive when accounting for fertilizer and fossil-fuel use in farm operations and milling.

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flowchart LR

RiceField[Rice Paddy Field] --> Water[Large Water Use]

RiceField --> Nfertilizer[Nitrogen Fertilizer]

Water --> CH4[Methane Emissions]

Nfertilizer --> N2O[Nitrous Oxide Emissions]

RiceField --> Straw[Crop Residue]

Straw --> CH4

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**\*\*Pests & Diseases.\*\*** Rice faces many biotic stresses. Notable **\*\*diseases\*\*** include rice blast (fungus *\*Magnaporthe oryzae\**), bacterial leaf blight (*\*Xanthomonas oryzae\**), sheath blight (*\*Rhizoctonia solani\**), rice tungro virus, and false smut (*\*Ustilaginoidea virens\**). **\*\*Insect pests\*\*** such as brown planthopper (BPH), stem borers, rice gall midge and green leafhoppers can devastate crops. Globally, pests and diseases are estimated to cause ~10–30% yield losses annually. Management strategies include planting resistant varieties, crop rotation/fallow, biological control (predators, parasitoids), judicious pesticide use, and field sanitation. For example, IRRI’s development of BPH- and virus-resistant varieties has averted previous major epidemics. Continued breeding for broad-spectrum resistance (using identified R genes) is a major research thrust.

**\*\*Climate Resilience & Adaptation.\*\*** Climate change (erratic monsoons, heat stress, salinity) threatens rice. Breeding for resilience is active: submergence-tolerant (Sub1) and drought-tolerant lines have been released in Asia. Salt-tolerant varieties (e.g. CSR-23 in India) help coastal areas. “Climate-smart” techniques (Alternate wetting and drying, system of rice intensification, stress-tolerant varieties) are promoted to cope with variability. Improved weather forecasting and insurance are also parts of adaptation strategies.

**\*\*Policy and Regulation.\*\*** Governments heavily influence rice through subsidies (for inputs or procurement), quality standards (milling quality, Basmati geographical indications), and trade policies (tariffs, export bans). For example, India’s 2023 export ban on non-basmati rice was aimed at capping domestic prices. Food aid programs (e.g. U.S. PL480) are major overseas rice suppliers in crises. Regional standards (ASEAN, EU) define maximum broken kernels and chalkiness. Rice fortification (with vitamins/minerals) is mandated or encouraged in some countries to combat malnutrition.

**\*\*Research Gaps and Future Directions.\*\*** Key challenges remain: raising productivity in Africa and rainfed Asia through improved agronomy; developing rice varieties that use water and nitrogen more efficiently and emit less methane; breeding for combined stresses (heat+salinity); and enhancing nutrition via biofortification or microbiota approaches. Precision agriculture (drones, sensors) and data analytics offer untapped potential to optimize fertilizer and water use. On the consumer side, shifting diets (e.g. to less refined carbohydrates) and processing innovations could reduce health risks. Continued research into pest/disease genomics, climate adaptation, and socioeconomic constraints (land tenure, credit) is needed to ensure rice food security in the face of changing climates and demographics.

**\*\*Tables & Figures\*\***

| **\*\*Variety Type\*\*** | **\*\*Subspecies/Origin\*\*** | **\*\*Typical Grain/Amylose\*\*** | **\*\*Main Uses/Regions\*\***  
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|                      |                                |                           |                                                     |
|----------------------|--------------------------------|---------------------------|-----------------------------------------------------|
|                      |                                |                           |                                                     |
| Indica               | O. sativa indica               | Long, high amylose        | Staple rice in South/Southeast Asia, USA long-grain |
| Japonica (temperate) | O. sativa japonica (temperate) | Short, sticky/low amylose | East Asian table rice (Japan, Korea)                |
| Japonica (tropical)  | O. sativa japonica (tropical)  | Medium, moderate amylose  | Upland rice, some Southeast Asia                    |
| Aromatic             | Varied (basmati, jasmine)      | Long, fragrant            | Biryani/pilaf (India, Pakistan) and Thai cuisine    |
| Glutinous (sticky)   | Mainly japonica                | Short, near-zero amylose  | East/Southeast Asian desserts (mochi, sticky rice)  |
| Hybrid (F1)          | Indica×Japonica                | Long, vigorous            | High-yield commercial cultivars (China, Bangladesh) |

|                        |                                        |                         |  |
|------------------------|----------------------------------------|-------------------------|--|
| <b>**Country**</b>     | <b>**2022/23 Production (milled)**</b> | <b>**Yield (t/ha)**</b> |  |
| -----                  | -----                                  | -----                   |  |
| <b>**China**</b>       | ~146 million tonnes                    | 6–7 (irrigated)         |  |
| <b>**India**</b>       | ~136 million tonnes                    | ~3.0 (avg)              |  |
| <b>**Indonesia**</b>   | ~55 million tonnes                     | ~5.3                    |  |
| <b>**Bangladesh**</b>  | ~50 million tonnes                     | ~4.5                    |  |
| <b>**Vietnam**</b>     | ~45 million tonnes                     | ~6.5                    |  |
| <b>*World Average*</b> | ~540 million tonnes (2025)             | ~4.0                    |  |

|                             |                              |                                             |  |
|-----------------------------|------------------------------|---------------------------------------------|--|
| <b>**Processing Stage**</b> | <b>**Operation**</b>         | <b>**Effect on Grain**</b>                  |  |
| -----                       | -----                        | -----                                       |  |
| Threshing                   | Separate grain from panicles | Paddy harvested                             |  |
| Drying                      | Reduce moisture to ~14%      | Prevents spoilage                           |  |
| Hulling (dehusking)         | Remove inedible hull         | Yields brown rice (bran on)                 |  |
| Parboiling (if used)        | Soak, steam, dry paddy       | Drives nutrients into grain, hardens grain  |  |
| Milling/Polishing           | Remove bran and germ         | Produces white rice (lowers fiber/vitamins) |  |
| Grading                     | Sort by size/color           | Classifies grades (broken, head rice)       |  |

Sources for tables: FAO and USDA production data, FAO yield statistics, and established agronomic references (IRRI/FAO manuals).

#### Key References:

FAO Agricultural Production Statistics 2000–2020 (FAOSTAT analytical brief)

USDA Rice Outlook (ERS, Dec 2025)

USDA/ERS Rice Trade Profiles

IFPRI blog on India rice exports (2023)

FAO International Rice Commission (IRC) report on nutrition

Frontiers Plant Science editorial on rice pests/diseases (2023)

FAO newsroom and reports on methane emissions in rice

Cornell University news on rice genetics (basmati)